

Pyruvate Kinase Deficiency: from a *PKLR* variant to haemolytic anaemia

Reading version of the project as committed in the GitHub repository (commit 082ff9b, 09 October 2026, 21:35: "Delete PKD_2_Overview_and_Team"), including the corrections prepared for commit (role swap of Bruno and Leo; status of 9 October), generated on 9 October 2026. The text is the committed text; only the layout differs. The reading guide at the end is not part of the repository.

Project plan: Gantt chart and risk analysis

Aarón Godino Martínez, Bruno Carrasquilla, Ettore Colombo, Leo Nader Al Hourani, Pol Monllau
Group 11DP-C · BSc in Bioinformatics (UPC-UB-UAB-UPF), Barcelona, Spain
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Introduction

Who we are

We are **Group 11DP-C**: Aarón Godino Martínez, Bruno Carrasquilla, Ettore Colombo, Leo Nader Al Hourani, and Pol Monllau, five first-year students of the BSc in Bioinformatics (UPC-UB-UAB-UPF). In the Disease Project of *Introduction to Bioinformatics*, each group is assigned a disease and investigates its molecular basis by analysing the sequences of the gene responsible and of the protein it encodes, and the three-dimensional structure of that protein. Our assigned disease is **Pyruvate Kinase Deficiency (PKD)**.

This document is our project plan: it defines our objectives, the work needed to reach them, who is responsible for each task, the schedule, and the risks we foresee.

Pyruvate Kinase Deficiency (PKD) in brief

The enzyme. Pyruvate kinase (PK) catalyses the last step of glycolysis: the irreversible transfer of a phosphate group from phosphoenolpyruvate to adenosine diphosphate (ADP), which yields pyruvate and adenosine triphosphate (ATP). Its activity is regulated allosterically (Bianchi and Fermo 2020). The enzyme is a tetramer of four identical subunits, and both of its human isoforms are activated by fructose 1,6-bisphosphate, an intermediate of an earlier step of glycolysis (Valentini *et al.* 2002).

Why red blood cells are affected. Mature red blood cells lack a nucleus and organelles, including mitochondria, and therefore depend entirely on glycolysis for the ATP they need to maintain their integrity and function (Bianchi and Fermo 2020, Fattizzo *et al.* 2022, Luke *et al.* 2023). When PK activity is insufficient, ATP levels fall and the upstream metabolite 2,3-diphosphoglycerate (2,3-DPG) accumulates; the red cells become rigid and less flexible and are removed prematurely from the circulation, mainly in the spleen (Bianchi and Fermo 2020, Fattizzo *et al.* 2022). Because the anaemia is chronic and 2,3-DPG favours the release of oxygen to the tissues, patients may tolerate it better than other forms of anaemia (Fattizzo *et al.* 2022). The result is a chronic non-spherocytic haemolytic anaemia, whose severity ranges from mild or fully

compensated haemolysis to life-threatening neonatal anaemia that requires exchange transfusion (Bianchi and Fermo 2020). In the 254 patients of the PKD Natural History Study, the most frequent complications were iron overload (48%) and gallstones (45%); pulmonary hypertension, extramedullary haematopoiesis, and leg ulcers were also observed (Grace *et al.* 2018).

The gene. PKD is caused by pathogenic variants in *PKLR*, located at 1q22 (HGNC 2026). The gene comprises 12 exons. Two tissue-specific promoters produce two mRNAs that differ only in their first exon: exon 1 is specific to the mRNA of the **R-type** isoform, present in red blood cells, and exon 2 to that of the **L-type** isoform, expressed in the liver; the other 10 exons are shared (Kanno *et al.* 1992, Bianchi and Fermo 2020). The R-type isoform has 574 amino acids and the L-type 543. The two differ only in their N-terminal region, where the first 33 residues of the R-type are replaced by two residues in the L-type; the same residue therefore has a number 31 positions lower in the L-type than in the R-type (UniProt Consortium 2026). Inheritance is autosomal recessive (Bianchi and Fermo 2020): affected individuals carry a pathogenic variant on both alleles of *PKLR*, either the same variant (homozygous) or two different ones (compound heterozygous). In the PKD Natural History Study, 142 of 177 unrelated patients (80%) were compound heterozygous, and 127 different pathogenic variants were identified, 84 of them missense (Bianchi *et al.* 2020); more than 300 different variants have been reported overall (Bianchi and Fermo 2020).

Frequency and treatment. PKD is the most common glycolytic enzyme defect associated with congenital non-spherocytic haemolytic anaemia (Bianchi and Fermo 2020). A systematic review estimated the prevalence of clinically diagnosed PKD in Western populations at 3.2–8.5 cases per million, and that of diagnosed and undiagnosed cases together at up to 51 per million, a figure extrapolated from allele frequencies (Secrest *et al.* 2020). Treatment has long been supportive: transfusions, splenectomy, and iron chelation (Grace and Barcellini 2020). Two disease-modifying treatments have since emerged: mitapivat, an oral allosteric activator of PK now approved for adults with PKD, and lentiviral gene therapy, which is being evaluated in clinical trials (Luke *et al.* 2023). In a phase 3 trial in adults not receiving regular transfusions, mitapivat increased haemoglobin levels compared with placebo (Al-Samkari *et al.* 2022).

The data we were given

Our instructors provided four files in FASTA format:

File	Content	Length
PKD_CDS_WT.fasta	coding sequence, wild type	1725 nucleotides
PKD_CDS_mutant.fasta	coding sequence, mutant	1725 nucleotides
PKD_protein_WT.fasta	protein sequence, wild type	574 amino acids
PKD_protein_mutant.fasta	protein sequence, mutant	574 amino acids

The headers (for example, `>PKD_CDS_WT`) state only the disease, the type of sequence, and whether the sequence is wild type or mutant. They contain no identifier for the gene, the isoform, or the variant; establishing these is our first task.

Key concepts

FASTA format. A plain-text format for biological sequences, named after the FASTA sequence-comparison programs (Pearson and Lipman 1988): a header line beginning with `>`, followed by the sequence written with one letter per nucleotide or amino acid. It is the standard input format of most bioinformatics tools.

Coding sequence and protein sequence. The coding sequence (CDS) is the part of a gene's transcript, from the start codon to the stop codon, that is translated into protein; it is usually written as a DNA sequence. Translating it codon by codon (one codon is three consecutive nucleotides) gives the amino acid sequence of the protein. Having both sequences allows us to describe a variant twice: as a change in the coding DNA and as its consequence for the amino acid sequence. The nomenclature of the Human Genome Variation Society (HGVS) marks the first with the prefix *c.* and the second with *p.* (den Dunnen *et al.* 2016).

Wild type and mutant. The wild-type sequence is the reference sequence; the mutant sequence carries a disease-associated variant. Comparing the two establishes the position and the nature of the variant.

Protein structure. Experimentally determined three-dimensional structures of proteins are deposited in the Protein Data Bank (PDB) (Berman *et al.* 2000). For human R-type PK, the structures of the wild-type enzyme and of disease-associated mutants have been determined, which makes it possible to relate each amino acid substitution to its position relative to the active site, the allosteric site, and the interfaces between domains and between subunits (Valentini *et al.* 2002).

Orthologs and conservation. Orthologs are genes in different species that descend from a single gene of their last common ancestor and diverged through speciation (Koonin 2005). Aligning the sequences of orthologous proteins shows which positions have been conserved during evolution. A residue conserved over long evolutionary distances is likely to be under structural or functional constraint, and sequence conservation is one of the most widely used predictors of functionally important residues (Capra and Singh 2007).

Our question and objectives

The Disease Project asks every group to answer the same question: *how can changes in gene/protein sequence and structure explain the molecular basis of disease, and what can comparison tell us about the affected biological function?*

Our general objective is to **explain** how the variant present in our mutant sequences changes the *PKLR* coding sequence, the sequence, structure, and function of the PK protein, and how these changes cause PKD; and to establish what comparing human PK with its orthologs in other species reveals about the affected function. We have set four specific objectives, one for each topic of the Disease Project Overview (IB Teaching Staff 2026a):

1. **Identify** the cause of the disease from the gene, protein, structure, and function perspectives.
2. **Describe** the differences between the wild-type and the mutant gene, protein, structure, and function.
3. **Explain** how these differences affect the patient.
4. **Compare** the gene, protein, structure, and function of human PK with those of its orthologs in at least two other species, including the rat, and infer what the comparison reveals about the affected function.

Our approach

The work is organized in six steps, each with a defined output; most of them apply methods taught in a specific class session.

Step	Work	Main sources and tools	Class session	Objective
1. Sequence characterization	Verify that each CDS translates into the corresponding protein sequence; identify and name every difference between wild type and mutant, in the coding sequence and in the protein sequence	Sequence translation and pairwise comparison	Data sources	1, 2
2. Background review	Review the normal function of the gene and the enzyme, and the mechanism, inheritance, and clinical features of PKD	PubMed, NCBI Gene, OMIM, UniProt	Data sources; paper structure and citation	1, 3
3. Comparative analysis	Retrieve the orthologous sequences, align them with the human sequence, and assess the conservation of the substituted position	UniProt, Ensembl, Clustal Omega	Sequence alignment (8 and 15 Oct)	4
4. Structural analysis	Locate the substituted residue in experimental structures, relative to the active and allosteric sites; superpose the wild-type and mutant structures	PDB, molecular visualization software	PDB (1 Oct); structure superposition (22 Oct)	1, 2
5. Pathogenicity assessment	Combine clinical classification, population frequency, conservation, structural, and biochemical data, distinguishing measured results from hypotheses and open questions	ClinVar, gnomAD, primary literature	Integrated bioinformatics reasoning (19 Oct)	2, 3
6. Synthesis and report	Write the answer to the project question as a manuscript in the <i>BMC Bioinformatics</i> format	GitHub, Markdown	Project management and GitHub	all

These steps are grouped into four work packages: **WP1**, project management and GitHub, for the whole duration of the project; **WP2**, data sources, literature, and sequences (steps 1 and 2); **WP3**, sequence and structure analysis (steps 3 and 4); and **WP4**, report, delivery, and defence (steps 5 and 6, and the preparation of the oral defence). The tasks are distributed among the five members so that each has a similar workload (10 to 12 working days of planned effort; Section 1), each task has a single owner, and each member commits their own work, so that the history of the repository records individual contributions.

Species compared

The comparison must include the rat (*Rattus norvegicus*), as specified in our disease assignment. The rat is also the species in which the production of both PK isoforms from a single gene — the L-type, expressed in the liver, and the R-type, present in red blood cells — was demonstrated (Noguchi *et al.* 1987), and experimental structures of its L-type isoform are available (Gassaway *et al.* 2019).

As the second species we propose the mouse (*Mus musculus*). Like those of human and rat, its PK sequence is a manually reviewed UniProt entry (UniProt Consortium 2025), and mice of the CBA/N strain carrying a spontaneous mutation at the *Pk-1* locus (the former symbol of *Pklr*) develop a non-spherocytic haemolytic anaemia, as in the human disease (Morimoto *et al.* 1995). Because rat and mouse are closely related rodents, comparing them shows whether a difference between the rat and human sequences is specific to the rat lineage. To infer which residue is ancestral, we will include the dog (*Canis lupus familiaris*) in the alignment as an outgroup: the dog lineage separated from the common ancestor of primates and rodents about 89 million years ago, before primates and rodents separated from each other, about 84 million years ago (Kumar *et al.* 2022).

Deliverables and schedule

The project is developed in a **GitHub repository**, to which each member commits their own contributions as the work progresses. At the end, the repository must contain the final **report** (a Markdown manuscript in the *BMC Bioinformatics* format, of at most 7 pages or 5000 words, whichever is shorter, with at most five figures or tables), a live **Project Status document**, this **Gantt chart** (the version frozen on 9 October and a live version updated until the end of the project), and all the **references** used (IB Teaching Staff 2026b, 2026c).

The project runs from 17 September 2026, when it was presented in class and our repository was created, to the oral defence on 5 November, Monday to Friday, excluding the holidays of 24 September (local) and 12 October (national). The key dates are:

- **9 October:** submission of this plan (18:00 in the Disease Project Overview, 22:00 on the course's Moodle page; we work to the earlier time);
- **22 October:** discussion of our Project Status document in class;
- **28 October, 18:00:** delivery of the final report on GitHub (the Overview gives "Monday October 28th", but 28 October 2026 is a Wednesday, so we have asked the instructors to confirm the date);
- **5 November:** oral defence.

Risks

We identified our risks from the requirements stated in the course instructions. The five scored as high are committing work in bulk or too late, an unequal contribution among members, selecting an incorrect database record or misinterpreting a result, developing the parts of the project in isolation, and failing to meet the report format or the delivery deadline. Each risk has an owner and a contingency plan.

Structure of this document

Section 1 details the objectives, the milestones, and the team; Section 2 presents the Gantt chart and the task specifications; Section 3 presents the risk analysis. References follow the Mini Oxford style required by the *BMC Bioinformatics* template.

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1. Project overview and team

Project name: Pyruvate Kinase Deficiency: from a *PKLR* variant to haemolytic anaemia

General objective

To **explain** how the variant present in mutant sequences changes the *PKLR* coding sequence and the sequence, structure, and function of the PK protein, how these changes cause Pyruvate Kinase Deficiency, and what comparing human PK with its orthologs in other species reveals about the affected function.

Specific objectives

Each specific objective corresponds to one topic of the Disease Project Overview and states the result that shows it has been reached.

ID	Objective	Accomplished when	Due
SO1	Identify the cause of the disease from the gene, protein, structure, and function perspectives.	The gene, isoform, and variant are identified and named in HGVS nomenclature at DNA and protein level , and the variant's clinical classification, population frequency, and published functional effect are recorded with sources.	Fri 16 Oct
SO2	Describe the differences between the wild-type and the mutant gene, protein, structure, and function.	The substituted residue is located relative to the active site, the allosteric site, and the interfaces; wild-type and mutant structures are superposed and their differences quantified; the functional difference is reported from the literature.	Mon 19 Oct
SO3	Explain how these differences affect the patient.	A causal chain from the variant to enzyme activity, red-cell ATP, haemolysis, and clinical severity is written, with each step supported by a source , and the evidence is classified as measured, hypothetical, or unknown.	Tue 20 Oct
SO4	Compare the gene, protein, structure, and function of human PK with those of its orthologs in at least two other species, including the rat.	Human PK is aligned with the rat and mouse orthologs and with the dog ortholog as outgroup; the residue at the substituted position is reported for every species and the lineage of each difference is stated; human and rat structures are compared around that position.	Tue 20 Oct

How we will reach each objective

ID	Overview topic	Class sessions applied	Tasks	Output in the report
SO1	Cause of the disease	Data sources; paper structure and citation	2.1, 2.2, 2.3, 2.4, 2.8	Background; Table 1 (variant and evidence)
SO2	Healthy versus disease state	Structural biology; PDB; structure superposition	2.1, 2.6, 3.3, 3.4	Results; Figure 2 (wild-type and mutant structures)
SO3	Impact on the patient	Integrated bioinformatics reasoning	2.4, 3.6, 3.7	Discussion; Figure 3 (from variant to patient)
SO4	Comparison with other species	Genomics; sequence alignment labs 1 and 2	2.5, 3.1, 3.2, 3.5, 3.9	Results; Table 2 (identity) and Figure 1 (alignment)

Planned outputs: two tables and three figures, the maximum of five outputs allowed by the template.

Milestones

ID	Milestone	Date	Source
M0	Project presented; repository created	Thu 17 Sep	Overview; commits
M1	Project plan submitted	Fri 9 Oct	Overview (18:00)
M2	Data collected (variant, sequences, structures, literature)	Mon 19 Oct	after 2.7
M3	Data analysis completed	Wed 21 Oct	after 3.8
M4	Project status document discussed	Thu 22 Oct	Overview; Formal Requirements
M5	Report draft complete	Mon 26 Oct	after 4.6
M6	Final report delivered	Wed 28 Oct	Overview (18:00)
M7	Oral defence (all members)	Thu 5 Nov	Course schedule

Project members and roles

Roles agreed provisionally; to be confirmed by the team.

Member	Role	Tasks owned
Aarón Godino Martínez	Product Owner	1.2, 2.4, 3.6, 3.7, 4.1, 4.6, 4.9, 4.10
Bruno Carrasquilla	Sequence Analyst	1.3, 2.1, 3.1, 3.2, 4.3, 4.7, 4.13, 4.16
Ettore Colombo	Project Manager	1.1, 1.6, 1.7, 1.8, 1.9, 2.8, 3.8, 4.5, 4.12
Leo Nader Al Hourani	Data and GitHub Manager	1.5, 2.2, 2.3, 2.5, 2.7, 3.9, 4.2, 4.8, 4.11, 4.15
Pol Monllau	Structure Analyst	1.4, 2.6, 3.3, 3.4, 3.5, 4.4, 4.14

Aarón Godino Martínez — Product Owner

First-year student of the BSc in Bioinformatics (UPC-UB-UAB-UPF). Past student of Cassablanca high school at Sabadell north and Castellarnau high school at Castellarnau; basic coding skills in python.

Responsibilities. Keeps the work focused on the project question; reviews the literature; writes the Background and the Abstract; assembles and delivers the report.

Bruno Carrasquilla — Sequence Analyst

First-year student of the BSc in Bioinformatics (UPC-UB-UAB-UPF). Student from Sant Ignasi Sarrià high school.

Responsibilities. Characterizes the supplied sequences; performs the pairwise and multiple alignments; writes the sequence Results; coordinates the read-through and the defence rehearsal.

Ettore Colombo — Project Manager

First-year student of the BSc in Bioinformatics (UPC-UB-UAB-UPF). BSc and MSc in Clinical Psychology. Currently conducting the thesis-related lab research for his MSc in Neuroscience (UAB). Basic coding skills in Python and R. Former professional actor. Primary caregiver since 2020.

Responsibilities. Plans and monitors the schedule; keeps the Project Status document; chairs the team meetings; writes the Discussion and Conclusions.

Leo Nader Al Hourani — Data and GitHub Manager

First-year student of the BSc in Bioinformatics (UPC-UB-UAB-UPF). Student from Salvador Espriu's high school

Responsibilities. Maintains the repository structure; retrieves identifiers, sequences, and variant evidence; archives the sources and the reference list; writes the Methods; checks the format; assembles the defence presentation.

Pol Monllau – Structure Analyst

First-year student of the BSc in Bioinformatics (UPC-UB-UAB-UPF). Student from Betania-Patmos

Responsibilities. Selects the structures; maps the substituted residue; performs the superpositions; writes the structure Results and prepares the figures.

Distribution of the planned work

Every member owns at least one task in each work package, so that each of us practises every phase of the project. All five members present at the oral defence: each prepares the slides of their own part (tasks 4.10 to 4.14).

Member	Tasks owned	Working days of effort	WP1	WP2	WP3	WP4
Aarón Godino Martínez	8	11	1.2	2.4	3.6, 3.7	4.1, 4.6, 4.9, 4.10
Bruno Carrasquilla	8	11	1.3	2.1	3.1, 3.2	4.3, 4.7, 4.13, 4.16
Ettore Colombo	9	11	1.1, 1.6, 1.7, 1.8, 1.9	2.8	3.8	4.5, 4.12
Leo Nader Al Hourani	10	12	1.5	2.2, 2.3, 2.5, 2.7	3.9	4.2, 4.8, 4.11, 4.15
Pol Monllau	7	10	1.4	2.6	3.3, 3.4, 3.5	4.4, 4.14

Effort counts the working days of each owned task; the weekly status update (1.7) counts as two days, one per meeting. Tasks completed before 9 October (Phase 0, Section 2) are not included.

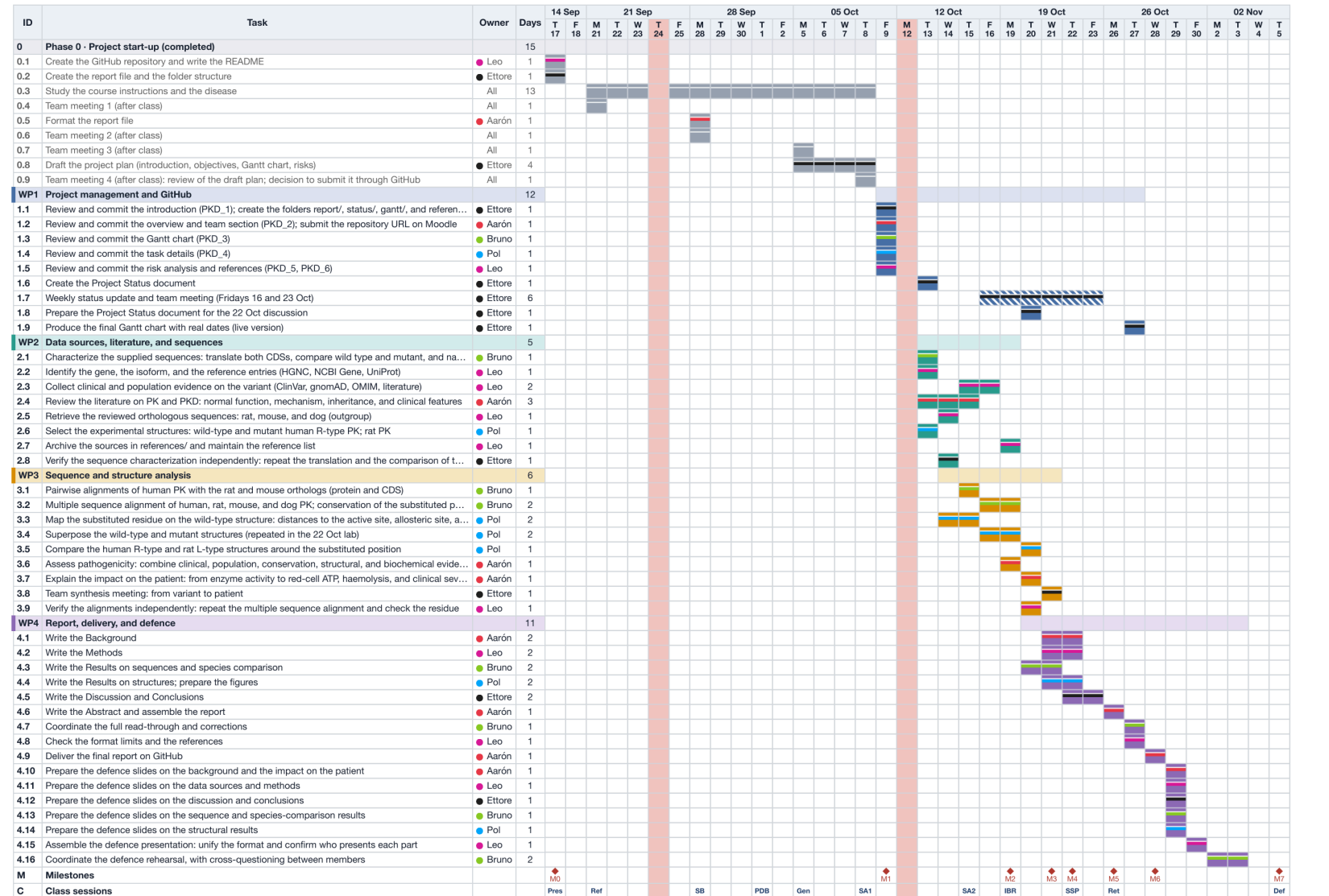
2. Gantt chart

The project runs from **Thu 17 Sep** to **Thu 5 Nov** (the oral defence), 34 working days (Monday to Friday, excluding the holidays of 24 September and 12 October). It has two parts: **Phase 0**, completed before this plan was submitted, and the **planned work** (WP1 to WP4) from 9 October. Each analysis is scheduled after the class session that teaches its method.

Work packages

WP	Name	Start	End	Working days	Status
Phase 0	Project start-up	Thu 17 Sep	Thu 8 Oct	15	Completed
WP1	Project management and GitHub	Fri 9 Oct	Tue 27 Oct	12	Planned
WP2	Data sources, literature, and sequences	Tue 13 Oct	Mon 19 Oct	5	Planned
WP3	Sequence and structure analysis	Wed 14 Oct	Wed 21 Oct	6	Planned
WP4	Report, delivery, and defence	Tue 20 Oct	Tue 3 Nov	11	Planned
Project		Thu 17 Sep	Thu 5 Nov	34	

Chart

11DP-C · Pyruvate Kinase Deficiency: from a *PKLR* variant to haemolytic anaemia · Gantt chart, plan of 9 October 2026

Phase (bar colour) Phase 0 - Project start-up (completed) WP1 - Project management and GitHub WP2 - Data sources, literature, and sequences WP3 - Sequence and structure analysis WP4 - Report, delivery, and defence Holiday Weekly checkpoints

Owner (line under the bar) Aarón Godino Martínez Bruno Carrasquilla Ettore Colombo Leo Nader Al Hourani Pol Monllau All members

Class sessions: Pres = Project presentation and access to GitHub · Ref = Paper structure, sources and references · SB = Structural biology · PDB = PDB lab · Gen = Genomics · SA1 = Sequence alignment lab 1 · SA2 = Sequence alignment lab 2 · IBR = Integrated bioinformatics reasoning · SSP = Structure superposition lab · Ret = Project retrospective (online, optional) · Def = Project defence. Days = working days. Weekends are omitted; holiday columns are shaded.

Bar colour shows the phase (grey: Phase 0, completed; WP1 to WP4 in the colours of the legend); the thin line under each bar shows its owner. Shaded columns are holidays; ♦ marks milestones; the bottom row shows the class sessions.

2. Gantt chart: task details

Days are working days. Predecessors must be finished before a task starts. Tasks 1.1 to 1.5 take place in sequence on 9 October: 1.1 creates the folders that the other four commits use.

ID	Task	Owner	Output	Days	Start	End	Predecessors	Accomplished when / Evidence
WP0	Phase 0 - Project start-up (completed)							
0.1	Create the GitHub repository and write the README	Leo	Repository 11DP-C_Pyruvate-Kinase-Deficiency_IB_2026	1	Thu 17 Sep	Thu 17 Sep	-	Commits of 17 Sep
0.2	Create the report file and the folder structure	Ettore	report.md, figures/	1	Thu 17 Sep	Thu 17 Sep	-	Commits of 17 Sep
0.3	Study the course instructions and the disease	All	Shared understanding of the task	15	Mon 21 Sep	Thu 8 Oct	-	Reported by the team
0.4	Team meeting 1 (after class)	All	Organization of the work	1	Mon 21 Sep	Mon 21 Sep	-	Reported by the team
0.5	Format the report file	Aarón	Formatted report.md	1	Mon 28 Sep	Mon 28 Sep	-	Commit of 28 Sep
0.6	Team meeting 2 (after class)	All	Organization of the work	1	Mon 28 Sep	Mon 28 Sep	-	Reported by the team
0.7	Team meeting 3 (after class)	All	Organization of the work	1	Mon 5 Oct	Mon 5 Oct	-	Reported by the team
0.8	Draft the project plan (introduction, objectives, Gantt chart, risks)	Ettore	Draft plan, sections PKD_1 to PKD_6	4	Mon 5 Oct	Thu 8 Oct	-	Draft files
0.9	Team meeting 4 (after class): review of the draft plan; decision to submit it through GitHub	All	Agreed plan and submission format	1	Thu 8 Oct	Thu 8 Oct	0.8	Reported by the team
WP1	Project management and GitHub							
1.1	Review and commit the introduction (PKD_1); create the folders report/, status/, gantt/, and references/	Ettore	Introduction and required folders in the repository	1	Fri 8 Oct	Fri 9 Oct	-	The four required folders exist and the introduction is committed by its reviewer.
1.2	Review and commit the overview and team section (PKD_2); submit the repository URL on Moodle	Aarón	PKD_2 committed; URL submitted	1	Fri 8 Oct	Fri 9 Oct	1.1	The URL is submitted on Moodle before 18:00 on 9 Oct.
1.3	Review and commit the Gantt chart (PKD_3)	Bruno	PKD_3 committed	1	Fri 8 Oct	Fri 9 Oct	1.1	The chart renders on GitHub and its dates match PKD_4.

ID	Task	Owner	Output	Days	Start	End	Predecessors	Accomplished when / Evidence
1.4	Review and commit the task details (PKD_4)	Pol	PKD_4 committed	1	Fri 8 Oct	Fri 9 Oct	1.1	Every task has an owner, output, duration, predecessor, and completion criterion.
1.5	Review and commit the risk analysis and references (PKD_5, PKD_6)	Leo	PKD_5 and PKD_6 committed	1	Fri 8 Oct	Fri 9 Oct	1.1	Each risk has a score and a contingency plan; every citation has a reference.
1.6	Create the Project Status document	Ettore	status/ document	1	Tue 13 Oct	Tue 13 Oct	1.2	It lists every task with owner, planned dates, real dates, status, and notes.
1.7	Weekly status update and team meeting (Fridays 16 and 23 Oct)	Ettore	Dated notes in status/	6	Fri 16 Oct	Fri 23 Oct	1.6	After each meeting, real dates, status changes, and incidents are recorded.
1.8	Prepare the Project Status document for the 22 Oct discussion	Ettore	Updated status/ document	1	Tue 20 Oct	Tue 20 Oct	1.6	Planned and real dates, status, and notes are current on 20 Oct.
1.9	Produce the final Gantt chart with real dates (live version)	Ettore	Final Gantt chart in gantt/	1	Tue 27 Oct	Tue 27 Oct	4.6	Every task shows its real dates, and deviations are explained.
WP2	Data sources, literature, and sequences							
2.1	Characterize the supplied sequences: translate both CDSs, compare wild type and mutant, and name the variant	Bruno	Variant table (DNA and protein level)	1	Tue 13 Oct	Tue 13 Oct	1.4	Every difference is listed at DNA and protein level and verified by a second member.
2.2	Identify the gene, the isoform, and the reference entries (HGNC, NCBI Gene, UniProt)	Leo	Table of identifiers	1	Tue 13 Oct	Tue 13 Oct	1.4	The gene symbol, isoform, and accession numbers are recorded with access dates.
2.3	Collect clinical and population evidence on the variant (ClinVar, gnomAD, OMIM, literature)	Leo	Evidence table with sources	2	Thu 15 Oct	Fri 16 Oct	2.1, 2.5	Classification, allele frequency, and published functional data are recorded with sources.
2.4	Review the literature on PK and PKD: normal function, mechanism, inheritance, and clinical features	Aarón	Annotated literature notes	3	Tue 13 Oct	Thu 15 Oct	1.2	Each statement planned for the Background has at least one primary or review source.
2.5	Retrieve the reviewed orthologous sequences: rat, mouse, and dog (outgroup)	Leo	FASTA files and accession list	1	Wed 14 Oct	Wed 14 Oct	2.2	Every sequence is saved in the repository with its accession number.
2.6	Select the experimental structures: wild-type and mutant human R-type PK; rat PK	Pol	Table of PDB entries (method, resolution, construct)	1	Tue 13 Oct	Tue 13 Oct	2.2	One wild-type and one mutant entry are chosen, with the reasons recorded.
2.7	Archive the sources in references/ and maintain the reference list	Leo	PDFs and Mini Oxford reference list	1	Mon 19 Oct	Mon 19 Oct	2.3, 2.4	Every cited source is archived and listed in the required style.

ID	Task	Owner	Output	Days	Start	End	Predecessors	Accomplished when / Evidence
2.8	Verify the sequence characterization independently: repeat the translation and the comparison of the four files	Ettore	Verification note	1	Wed 14 Oct	Wed 14 Oct	2.1	The variant table of task 2.1 is confirmed, or each discrepancy is resolved with its owner.
WP3	Sequence and structure analysis							
3.1	Pairwise alignments of human PK with the rat and mouse orthologs (protein and CDS)	Bruno	Alignments and identity table	1	Thu 15 Oct	Thu 15 Oct	2.1, 2.5	Percent identity and similarity are reported for each pair, with tool and parameters.
3.2	Multiple sequence alignment of human, rat, mouse, and dog PK; conservation of the substituted position and lineage	Bruno	Alignment figure and conservation notes	2	Fri 16 Oct	Mon 19 Oct	3.1	The residue at the substituted position is reported for every species, and lineage is stated.
3.3	Map the substituted residue on the wild-type structure: distances to the active site, allosteric site, and interfaces	Pol	Annotated structure figure and distance table	2	Wed 14 Oct	Thu 15 Oct	2.1, 2.6	Distances are measured in a stated structure and chain.
3.4	Superpose the wild-type and mutant structures (repeated in the 22 Oct lab)	Pol	Superposition figure and RMSD values	2	Fri 16 Oct	Mon 19 Oct	3.3	Global and local differences are quantified with the method stated.
3.5	Compare the human R-type and rat L-type structures around the substituted position	Pol	Comparison figure and notes	1	Tue 20 Oct	Tue 20 Oct	3.4, 3.2	The local environment of the position is described in both species.
3.6	Assess pathogenicity: combine clinical, population, conservation, structural, and biochemical evidence	Aarón	Evidence summary table	1	Mon 19 Oct	Mon 19 Oct	2.3, 2.4, 3.1	Each line of evidence is classified as measured result, hypothesis, or open question.
3.7	Explain the impact on the patient: from enzyme activity to red-cell ATP, haemolysis, and clinical severity	Aarón	One-page causal summary	1	Tue 20 Oct	Tue 20 Oct	3.6	Each step from variant to clinical picture is supported by a cited source or our own result.
3.8	Team synthesis meeting: from variant to patient	Ettore	Meeting notes	1	Wed 21 Oct	Wed 21 Oct	3.5, 3.7, 3.9	Every member can explain the full chain from variant to patient.
3.9	Verify the alignments independently: repeat the multiple sequence alignment and check the residue	Leo	Verification note	1	Tue 20 Oct	Tue 20 Oct	3.2	The residues reported in task 3.2 are confirmed, or each discrepancy is resolved with its owner.
WP4	Report, delivery, and defence							
4.1	Write the Background	Aarón	Background section in report/	2	Wed 21 Oct	Thu 22 Oct	2.4	It states the problem and the question, with every claim cited.
4.2	Write the Methods	Leo	Methods section in report/	2	Wed 21 Oct	Thu 22 Oct	2.7, 3.9	Another student could reproduce every analysis from it.

ID	Task	Owner	Output	Days	Start	End	Predecessors	Accomplished when / Evidence
4.3	Write the Results on sequences and species comparison	Bruno	Results, part 1	2	Tue 20 Oct	Wed 21 Oct	3.2	It reports what was measured, without interpretation, with numbered figures or tables.
4.4	Write the Results on structures; prepare the figures	Pol	Results, part 2; figures	2	Wed 21 Oct	Thu 22 Oct	3.5	Figures are numbered, captioned, and stored in the repository.
4.5	Write the Discussion and Conclusions	Ettore	Discussion and Conclusions	2	Thu 22 Oct	Fri 23 Oct	3.8	They answer the project question and state the limitations.
4.6	Write the Abstract and assemble the report	Aarón	Complete report	1	Mon 26 Oct	Mon 26 Oct	4.1, 4.2, 4.3, 4.4, 4.5	All BMC sections are present; the Abstract has at most 350 words.
4.7	Coordinate the full read-through and corrections	Bruno	Corrected report	1	Tue 27 Oct	Tue 27 Oct	4.6	Every member has read the whole report and all comments are resolved.
4.8	Check the format limits and the references	Leo	Completed checklist	1	Tue 27 Oct	Tue 27 Oct	4.6	At most 7 pages or 5 000 words and five figures or tables; references in Mini Oxford style.
4.9	Deliver the final report on GitHub	Aarón	Final repository	1	Wed 28 Oct	Wed 28 Oct	4.7, 4.8, 1.9	All files are committed by 12:00 on 28 Oct, six hours before the 18:00 deadline.
4.10	Prepare the defence slides on the background and the impact on the patient	Aarón	Slides, part 1	1	Thu 29 Oct	Thu 29 Oct	4.9	The slides cover the Background and SO3, and their presenter is named.
4.11	Prepare the defence slides on the data sources and methods	Leo	Slides, part 2	1	Thu 29 Oct	Thu 29 Oct	4.9	The slides cover the Methods, and their presenter is named.
4.12	Prepare the defence slides on the discussion and conclusions	Ettore	Slides, part 3	1	Thu 29 Oct	Thu 29 Oct	4.9	The slides answer the project question and state the limitations.
4.13	Prepare the defence slides on the sequence and species-comparison results	Bruno	Slides, part 4	1	Thu 29 Oct	Thu 29 Oct	4.9	The slides cover SO1 and SO4 results, and their presenter is named.
4.14	Prepare the defence slides on the structural results	Pol	Slides, part 5	1	Thu 29 Oct	Thu 29 Oct	4.9	The slides cover SO2 results, and their presenter is named.
4.15	Assemble the defence presentation: unify the format and confirm who presents each part	Leo	Complete presentation	1	Fri 30 Oct	Fri 30 Oct	4.10, 4.11, 4.12, 4.13, 4.14	One presentation in a single format, with every member presenting a part.
4.16	Coordinate the defence rehearsal, with cross-questioning between members	Bruno	Rehearsal and list of likely questions	2	Mon 2 Nov	Tue 3 Nov	4.15	Every member can answer questions on every part of the project.

3. Risk analysis

We identified our risks from the requirements of the course documents (Disease Project Overview, Formal Requirements, *BMC Bioinformatics* template, Use of GenAI in IB) and from the descriptors of the marking rubric, and we checked them against the current state of our repository on 8 October. Following the method taught in class, each risk is scored as **Risk = Likelihood × Impact**, both from 1 (low) to 3 (high): **1–2 Low, 3–5 Medium, 6–9 High**. Each risk has an owner, who monitors it and activates the contingency plan when needed.

Risk matrix

Likelihood ↓ / Impact →	1 (low)	2 (moderate)	3 (high)
3 (likely)	—	—	R1, R2 (9)
2 (possible)	—	R4, R6, R9 (4)	R5, R8, R10 (6)
1 (unlikely)	—	R7 (2)	R3 (3)

Risk register

ID	Risk	L	I	Score	Level	Owner	Status on 8 Oct	Source
R1	Commits made in bulk or near the deadlines	3	3	9	High	Ettore	Partly occurred: 11 commits, none since 28 Sep	Formal Requirements I; rubric, Project Documentation
R2	Unequal contribution among members	3	3	9	High	Pol	Partly occurred: three of five members have committed	Formal Requirements I; rubric, Project Documentation and Project Execution; Use of GenAI in IB
R3	Repository structure not compliant with the requirements	1	3	3	Medium	Bruno	Occurred: required folders missing; correction planned in task 1.1	Formal Requirements I–II; rubric, Project Documentation
R4	Project Status document or live Gantt chart not kept up to date	2	2	4	Medium	Ettore	Open	Formal Requirements III–IV; rubric, Project Scheduling
R5	Incorrect database record selected, or database result misinterpreted	2	3	6	High	Bruno	Open	Rubric, Bioinformatic database literacy
R6	Alignment or structural results misinterpreted	2	2	4	Medium	Pol	Open	Rubric, Bioinformatic analysis; Structural and evolutionary reasoning
R7	Species comparison insufficiently justified, or evolutionary changes inferred without an outgroup	1	2	2	Low	Leo	Open	Overview, Scope Expectations; rubric, Bioinformatic analysis
R8	Parts of the project developed in isolation; members aware only of their own tasks	2	3	6	High	Leo	Open	Rubric, Knowledge integration; Oral scientific communication
R9	Statements supported by weak or unverified sources, including unverified GenAI output	2	2	4	Medium	Aarón	Open	Rubric, Evidence gathering; Use of GenAI in IB

ID	Risk	L	I	Score	Level	Owner	Status on 8 Oct	Source
R10	Report format, length limits, or delivery deadline not met	2	3	6	High	Aarón	Open	Overview, Timelines; Formal Requirements I; <i>BMC Bioinformatics</i> template

Analysis and contingency plans

High risks

- **R1 (9).** Our identified risk is *committing our work in bulk or near the deadlines*, we have realized because the Formal Requirements state that a repository committed once or very few times before the deadline will be heavily penalized, and our history shows 11 commits made on two days (17 and 28 September) and none since; our contingency plan consists of committing the plan section by section on 9 October (tasks 1.1–1.5), committing at the end of every work session thereafter, and reviewing the commit history at each weekly meeting (task 1.7).
- **R2 (9).** Our identified risk is *an unequal contribution among members*, we have realized because the Formal Requirements state that contributions concentrated in one member severely penalize the whole team, only human commits are assessed (Use of GenAI in IB), and two of us have not yet committed; our contingency plan consists of an equal distribution of the planned work (seven tasks per member, Section 1), each member committing their own work from 9 October, recording each member's contributions in the Project Status document, and an Authors' contributions section in the report.
- **R5 (6).** Our identified risk is *selecting an incorrect database record or misinterpreting a database result*, we have realized because several records exist for the same object: *PKLR* encodes two isoforms, UniProt contains reviewed and unreviewed entries, the gene symbol (*PKLR*) differs from the UniProt entry name (*KPYR_HUMAN*), the PDB codes printed in Valentini *et al.* (2002) have been superseded, and only four *PKLR* entries in UniProt are manually reviewed (human, rat, mouse, and dog); our contingency plan consists of using reviewed entries only, which is why we compare the rat and the mouse and use the dog as outgroup, recording every accession number with its access date and the reason for its choice (tasks 2.2, 2.5, 2.6), and having a second member verify each record and the variant table (task 2.8).
- **R8 (6).** Our identified risk is *developing the parts of the project in isolation, so that each member understands only their own tasks*, we have realized because the rubric marks as unattained a project generated in a disconnected way and a defence showing "niche awareness"; our contingency plan consists of a synthesis meeting before writing (task 3.8), a full read-through of the report by every member (task 4.7), each member preparing and presenting their own part of the defence (tasks 4.10 to 4.14), and a rehearsal in which each member answers questions on the parts of others (task 4.16).
- **R10 (6).** Our identified risk is *failing to meet the report format, its length limits, or the delivery deadline*, we have realized because the report must follow the *BMC Bioinformatics* structure with at most 7 pages or 5 000 words and five figures or tables, and contributions committed after 18:00 on 28 October are discarded; our contingency plan consists of a word budget per section, a format and reference check on 27 October (task 4.8), assembling the report two days before the deadline (task 4.6), and committing the final version by 12:00 on 28 October, six hours before the deadline (task 4.9).

Medium risks

- **R3 (3).** Our identified risk is *a repository structure that does not comply with the requirements*, we have realized because the Formal Requirements require the folders `report/`, `status/`, `gantt/`, and `references/`, which our repository does not yet contain; our contingency plan consists of creating them on 9 October (task 1.1) and checking the structure at each weekly meeting (task 1.7).

- **R4 (4).** Our identified risk is *not keeping the Project Status document and the live Gantt chart up to date*, we have realized because both must be maintained until the end of the project and the rubric assesses the comparison between planned and real dates; our contingency plan consists of creating the status document on 13 October (task 1.6), updating both at each weekly meeting (task 1.7), and producing the final chart with real dates (task 1.9).
- **R6 (4).** Our identified risk is *misinterpreting alignment or structural results*, we have realized because we learn these methods during the project (labs on 8, 15, and 22 October) and the rubric marks incorrect interpretation as unattained; our contingency plan consists of scheduling each analysis after the lab that teaches it, asking the instructors during the labs, having a second member repeat the multiple alignment (task 3.9), and repeating the superposition in the 22 October lab (task 3.4).
- **R9 (4).** Our identified risk is *supporting statements with weak or unverified sources, including GenAI output*, we have realized because the rubric requires trusted, relevant sources deposited in the repository, and the GenAI guidelines make us liable for every error we include; our contingency plan consists of using primary articles for facts and reviews for context, verifying every statement suggested by GenAI against its original source, and archiving every cited source in `references/` (task 2.7).

Low risk

- **R7 (2).** Our identified risk is *an insufficiently justified species comparison, or evolutionary changes inferred without an outgroup*, we have realized because the rubric requires the chosen species to be justified and useful, and three sequences alone cannot show which residue is ancestral; our contingency plan consists of stating the reason for each species in the report, aligning the dog sequence as an outgroup (tasks 2.5, 3.2), and making claims about the lineage of a change only when the outgroup supports them.

Monitoring

The register is copied into the Project Status document (task 1.6). At each weekly meeting (task 1.7), the owner of each risk reports its status, any change in its score, the actions taken, and whether the contingency plan was activated.

References

All sources used in the project (Disease Project Formal Requirements, section II.4). PDFs are added to this folder as they are cited (task 2.7). Sources cited in the plan of 9 October:

Mini Oxford SCIMED style, as required by the BMC Bioinformatics template: alphabetical by first author; four or more authors given as the first author followed by et al.; page spans elided. Every entry was checked against its PubMed record or the source itself on 8–9 October 2026.

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Project Status

Live Project Status document of Group 11DP-C (Disease Project Formal Requirements, section III). It is generated from the *Project Status* sheet of [../gantt/Gantt live 11DP-C.xlsx](#), where each member updates the real dates, status, and notes of their own tasks.

Task tracking — Updated up to 09.10

Task	Resp.	Plan. Start Date	Plan. Finish Date	Real Start Date	Real Finish Date	Status	Notes
Phase 0 · Project start-up (completed)							
0.1 Create the GitHub repository and write the README	Leo	17.09	17.09	17.09	17.09	● Completed	Evidence: Commits of 17 Sep.
0.2 Create the report file and the folder structure	Ettore	17.09	17.09	17.09	17.09	● Completed	Evidence: Commits of 17 Sep.
0.3 Study the course instructions and the disease	All	21.09	08.10	21.09	08.10	● Completed	Evidence: Reported by the team.
0.4 Team meeting 1 (after class)	All	21.09	21.09	21.09	21.09	● Completed	Evidence: Reported by the team.
0.5 Format the report file	Aarón	28.09	28.09	28.09	28.09	● Completed	Evidence: Commit of 28 Sep.
0.6 Team meeting 2 (after class)	All	28.09	28.09	28.09	28.09	● Completed	Evidence: Reported by the team.
0.7 Team meeting 3 (after class)	All	05.10	05.10	05.10	05.10	● Completed	Evidence: Reported by the team.
0.8 Draft the project plan (introduction, objectives, Gantt chart, risks)	Ettore	05.10	08.10	05.10	08.10	● Completed	Evidence: Draft files.
0.9 Team meeting 4 (after class): review of the draft plan; decision to submit it through GitHub rather than as a PDF	All	08.10	08.10	08.10	08.10	● Completed	Evidence: Reported by the team.
WP1 · Project management and GitHub							

Task	Resp.	Plan. Start Date	Plan. Finish Date	Real Start Date	Real Finish Date	Status	Notes
1.1 Review and commit the introduction (PKD_1); create the folders report/, status/, gantt/, and references/	Ettore	09.10	09.10	09.10	09.10	● Completed	Introduction committed by Ettore on 9 Oct; the folders gantt/, status/, references/, and report/ were created by Leo and Bruno on 8 Oct.
1.2 Review and commit the overview and team section (PKD_2); submit the repository URL on Moodle	Aarón	09.10	09.10	08.10		● In progress	PKD_2 committed by Aarón on 8 Oct (roles of Bruno and Leo swapped by team decision); Moodle URL submission to be confirmed.
1.3 Review and commit the Gantt chart (PKD_3)	Bruno	09.10	09.10	08.10	08.10	● Completed	Committed by Leo on 8 Oct and edited by Pol; owner after the role swap: Bruno.
1.4 Review and commit the task details (PKD_4)	Pol	09.10	09.10	08.10	08.10	● Completed	Committed by Pol on 8 Oct; owner columns corrected for the role swap.
1.5 Review and commit the risk analysis and references (PKD_5, PKD_6)	Leo	09.10	09.10	08.10	09.10	● Completed	Committed by Bruno (PKD_6 on 8 Oct, PKD_5 on 9 Oct), references updated by Ettore on 9 Oct; owner after the role swap: Leo.
1.6 Create the Project Status document	Ettore	13.10	13.10			● Pending	
1.7 Weekly status update and team meeting (Fridays 16 and 23 Oct)	Ettore	16.10	23.10			● Pending	
1.8 Prepare the Project Status document for the 22 Oct discussion	Ettore	20.10	20.10			● Pending	
1.9 Produce the final Gantt chart with real dates (live version)	Ettore	27.10	27.10			● Pending	
WP2 · Data sources, literature, and sequences							
2.1 Characterize the supplied sequences: translate both CDSs, compare wild type and mutant, and name the variant in HGVS nomenclature	Bruno	13.10	13.10			● Pending	
2.2 Identify the gene, the isoform, and the reference entries (HGNC, NCBI Gene, UniProt)	Leo	13.10	13.10			● Pending	
2.3 Collect clinical and population evidence on the variant (ClinVar, gnomAD, OMIM, literature)	Leo	15.10	16.10			● Pending	
2.4 Review the literature on PK and PKD: normal function, mechanism, inheritance, and clinical features	Aarón	13.10	15.10			● Pending	
2.5 Retrieve the reviewed orthologous sequences: rat, mouse, and dog (outgroup)	Leo	14.10	14.10			● Pending	

Task	Resp.	Plan. Start Date	Plan. Finish Date	Real Start Date	Real Finish Date	Status	Notes
2.6 Select the experimental structures: wild-type and mutant human R-type PK; rat PK	Pol	13.10	13.10			 Pending	
2.7 Archive the sources in references/ and maintain the reference list	Leo	19.10	19.10			 Pending	
2.8 Verify the sequence characterization independently: repeat the translation and the comparison of the four files	Ettore	14.10	14.10			 Pending	
WP3 · Sequence and structure analysis							
3.1 Pairwise alignments of human PK with the rat and mouse orthologs (protein and CDS)	Bruno	15.10	15.10			 Pending	
3.2 Multiple sequence alignment of human, rat, mouse, and dog PK; conservation of the substituted position and lineage of each difference	Bruno	16.10	19.10			 Pending	
3.3 Map the substituted residue on the wild-type structure: distances to the active site, the allosteric site, and the domain and subunit interfaces	Pol	14.10	15.10			 Pending	
3.4 Superpose the wild-type and mutant structures (repeated in the 22 Oct lab)	Pol	16.10	19.10			 Pending	
3.5 Compare the human R-type and rat L-type structures around the substituted position	Pol	20.10	20.10			 Pending	
3.6 Assess pathogenicity: combine clinical, population, conservation, structural, and biochemical evidence	Aarón	19.10	19.10			 Pending	
3.7 Explain the impact on the patient: from enzyme activity to red-cell ATP, haemolysis, and clinical severity	Aarón	20.10	20.10			 Pending	
3.8 Team synthesis meeting: from variant to patient	Ettore	21.10	21.10			 Pending	
3.9 Verify the alignments independently: repeat the multiple sequence alignment and check the residue at the substituted position in every species	Leo	20.10	20.10			 Pending	
WP4 · Report, delivery, and defence							
4.1 Write the Background	Aarón	21.10	22.10			 Pending	
4.2 Write the Methods	Leo	21.10	22.10			 Pending	

Task	Resp.	Plan. Start Date	Plan. Finish Date	Real Start Date	Real Finish Date	Status	Notes
4.3 Write the Results on sequences and species comparison	Bruno	20.10	21.10			● Pending	
4.4 Write the Results on structures; prepare the figures	Pol	21.10	22.10			● Pending	
4.5 Write the Discussion and Conclusions	Ettore	22.10	23.10			● Pending	
4.6 Write the Abstract and assemble the report	Aarón	26.10	26.10			● Pending	
4.7 Coordinate the full read-through and corrections	Bruno	27.10	27.10			● Pending	
4.8 Check the format limits and the references	Leo	27.10	27.10			● Pending	
4.9 Deliver the final report on GitHub	Aarón	28.10	28.10			● Pending	
4.10 Prepare the defence slides on the background and the impact on the patient	Aarón	29.10	29.10			● Pending	
4.11 Prepare the defence slides on the data sources and methods	Leo	29.10	29.10			● Pending	
4.12 Prepare the defence slides on the discussion and conclusions	Ettore	29.10	29.10			● Pending	
4.13 Prepare the defence slides on the sequence and species-comparison results	Bruno	29.10	29.10			● Pending	
4.14 Prepare the defence slides on the structural results	Pol	29.10	29.10			● Pending	
4.15 Assemble the defence presentation: unify the format and confirm who presents each part	Leo	30.10	30.10			● Pending	
4.16 Coordinate the defence rehearsal, with cross-questioning between members	Bruno	02.11	03.11			● Pending	

Status values (Formal Requirements, section III): ● Completed · ● In progress · ● Incident · ● Pending · ● Backlog · ● Canceled. ⚠ = real date later than planned. No open task is past its planned finish on 09.10.

Reading guide: how this plan answers the assignment

Added in this reading version; not part of the repository. Requirements as listed in the Moodle task *Gantt Chart & Risk Analysis Submission* (due 9 October 2026).

Moodle section	Requirement	Where it is answered
1. Project Overview & Team	Project name	Section 1 (project name; also the title)
	General objective: one broad, overarching destination	Section 1, General objective
	Specific objectives: 3 to 5 narrow, measurable outcomes	Section 1, Specific objectives SO1–SO4, each with an 'accomplished when' criterion and a due date
	Milestones	Section 1, Milestones M0–M7
	Project members and roles: background, role, responsibilities	Section 1, Project members and roles
2. Gantt Chart Structure & Data	Work packages	Section 2, Work packages (Phase 0 and WP1–WP4)
	Tasks breakdown {WP}.{Task}	Section 2, task details (0.1–0.9, 1.1–4.16)
	For every task: owner, output deliverable, duration in working days, 'accomplished when', predecessors	Section 2, task details
	Calendar and dates of every task, WP, and the whole project; working days and local holidays	Section 2: chart, Work packages table, task details; holidays 24 Sep and 12 Oct
3. Risk Analysis	Risk assessment matrix, Risk = Likelihood × Impact (1–2 Low, 3–5 Medium, 6–9 High)	Section 3, Risk matrix and register
	For each risk: 'Our identified risk is..., we have realized because...; our contingency plan consists of...'	Section 3, Analysis and contingency plans